



Data science and database technology

Introduction to the course

Transaction processing

➤ On Line Transaction Processing (OLTP)

- Traditional DBMS usage

➤ Characterized by

- snapshot of current data values
- detailed data, relational representation
- structured, repetitive operations
- read/write access to few records
- short transactions
- isolation, reliability, and integrity are critical (ACID)
- database size $\approx$ 100MB-GB

Analytical processing

➤ On Line Analytical Processing (OLAP)

- Decision support applications

➤ Characterized by

- “historical” data
- consolidated, integrated data
- ad hoc applications
- read access to millions of records
- complex queries
- consistency before and after periodical loads
- database size $\approx$ 100GB-TB

Course content

➤ First part (weeks 1-7)

- Data warehouse design
- OLAP analysis
- Data science and data mining

➤ Second part (weeks 8-14)

- DBMS server technology
- Distributed databases
- NoSQL databases (MongoDB, Elastic)

Course structure

- The course includes
 - lessons
 - classroom exercises
 - laboratories
- Laboratory sessions propose experimental activities on the most widespread commercial and open-source products
 - Students are partitioned in two groups
 - Lab sessions will start on the fourth week

➤ Course books

- Golfarelli, Rizzi, 'Data Warehouse Design: modern principles and methodologies', McGraw Hill, 2021
- Tan, Steinbach, Kumar, *Introduction to data mining*, Pearson, 2006
- Atzeni, Ceri, Fraternali, Paraboschi, Torlone, 'Basi di dati', 5 ed., McGraw Hill, 2018.
- Dan Sullivan, *NoSQL for Mere Mortals*, Addison-Wesley Professional, 2015
- Kristina Chodorow, Shannon Bradshaw, *MongoDB: The Definitive Guide (Powerful and Scalable Data Storage)*, 3 ed. O'Reilly Media, 2018
- Gormley, Tong, *Elastic Search: The Definitive Guide*, O'Reilly, 2015

➤ Other books

- Ramakrishnan, Gehrke, *Database Management Systems*, McGraw-Hill, 2004
- Kimball e altri, *several books and white papers on data warehouse design methodologies and case studies*, Wiley
- Han, Kamber, *Data mining: concepts and techniques*, Morgan Kaufmann, 2006

Assessment and grading criteria

➤ Exam

- Written test (mandatory)
 - A set of design exercises
 - A set of theory questions and exercises
 - Textbooks, notes, electronic devices of any kind are *not allowed* during the written part
- Individual project on the main topics of the lectures (optional)
 - homework assigned during the course
 - homework must be delivered at predefined deadlines during the course

➤ Further details about exam structure and grading criteria are available on the didactic portal